TradeCraft OS

The sovereign operating system for the modern trade campus.

Grading, attendance, curriculum, tutoring and support — designed by a master tradesman and systems architect, running on your own hardware.

Journeyman Electrician & EPA Certified | **Runs on-campus** | **Works with the internet off** | **Live at Ohio Business College**

Michael Gorby Jr. • Journeyman Electrician • EPA Certified • Founder & Lead Architect

EXECUTIVE BRIEFING • BOARD OF DIRECTORS

01 / THE PROBLEM

Instructors were hired to teach a trade — not to be data-entry clerks.

Paperwork paralysis

Stacks of exams and lab sheets get graded by hand, night after night.

Typing it twice

Scores are re-keyed by hand into the school's official gradebook.

Waiting on support

When a tool breaks, class stalls until someone with a ticket system answers.

Cloud fragility

One internet outage and the entire classroom loses its tools.

In plain English: When you hire a master tradesman to teach apprentices, every hour they spend fighting clunky cloud software is an hour stolen from teaching the craft.

TRADECRAFT

One system. Twelve engines.

Everything a trade campus runs on — from the day a student enrolls to the day they graduate.

SOS Action Bot
Type what you need. It does it.

Scan-to-Grade
One-touch printer scan, graded.

DocuAI
Drag-and-drop grading & documents.

Gradebook Sync
Pushes grades to the school's system.

Attendance
Clock hours, audit-ready.

Curriculum Vault
Any trade. Any dataset.

Study Buddy
A personal tutor, 24/7.

TradeClash
Learning games from your curriculum.

Testing Grounds
Secure, proctored exams.

Lab Builder
Labs & worksheets in seconds.

Instant Reports
Any report, in milliseconds.

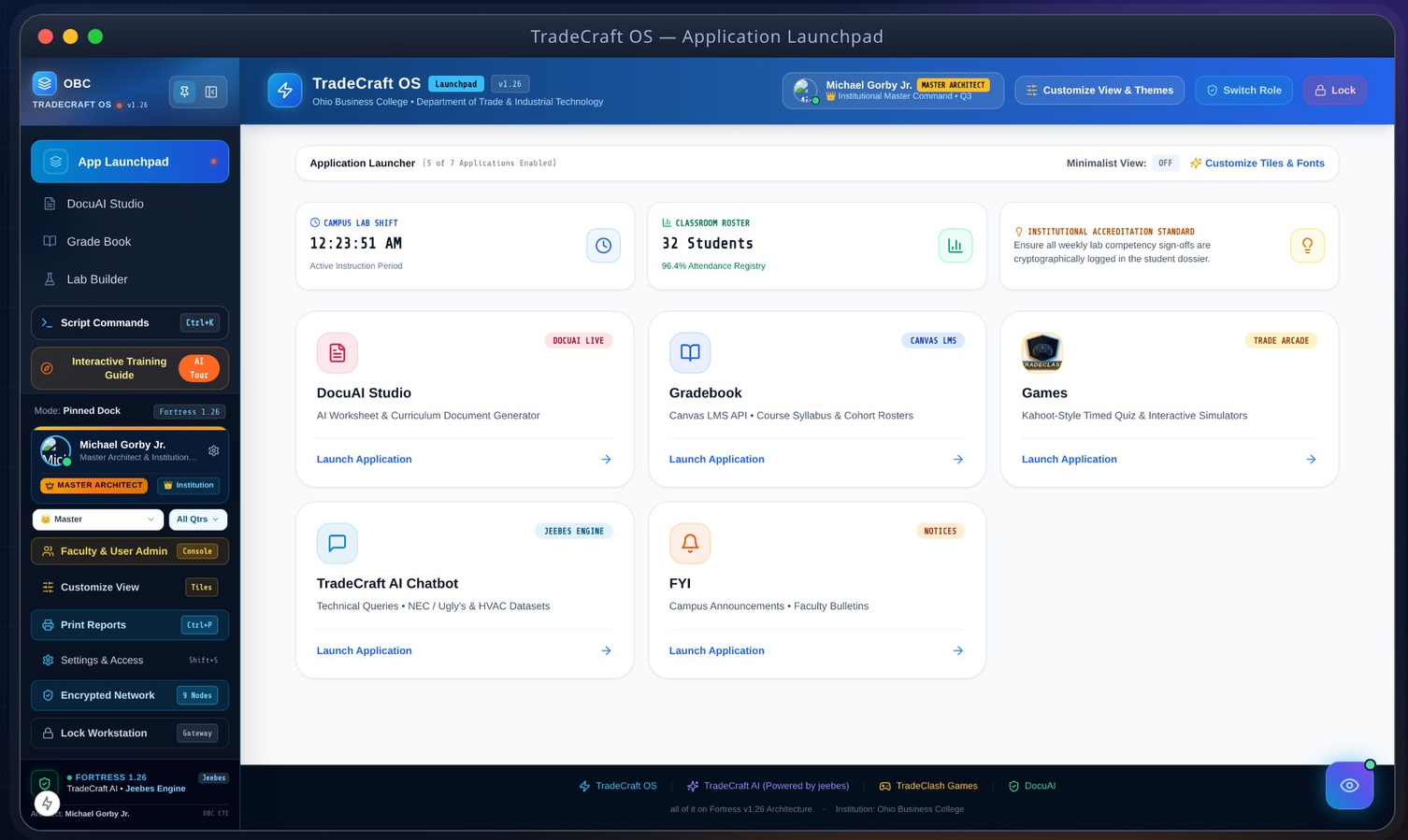
Redundant Core
Local machines. Never down.

One screen. Every tool.

Big, labeled buttons. No menu maze. No manuals.

- Gradebook, DocuAI, games, chatbot, notices — one tap each
- Personalized per user: tiles, themes, fonts, role
- Sign in once, work everywhere
- Built-in AI tour walks new users through it

In plain English: If you can use a phone, you can use TradeCraft.



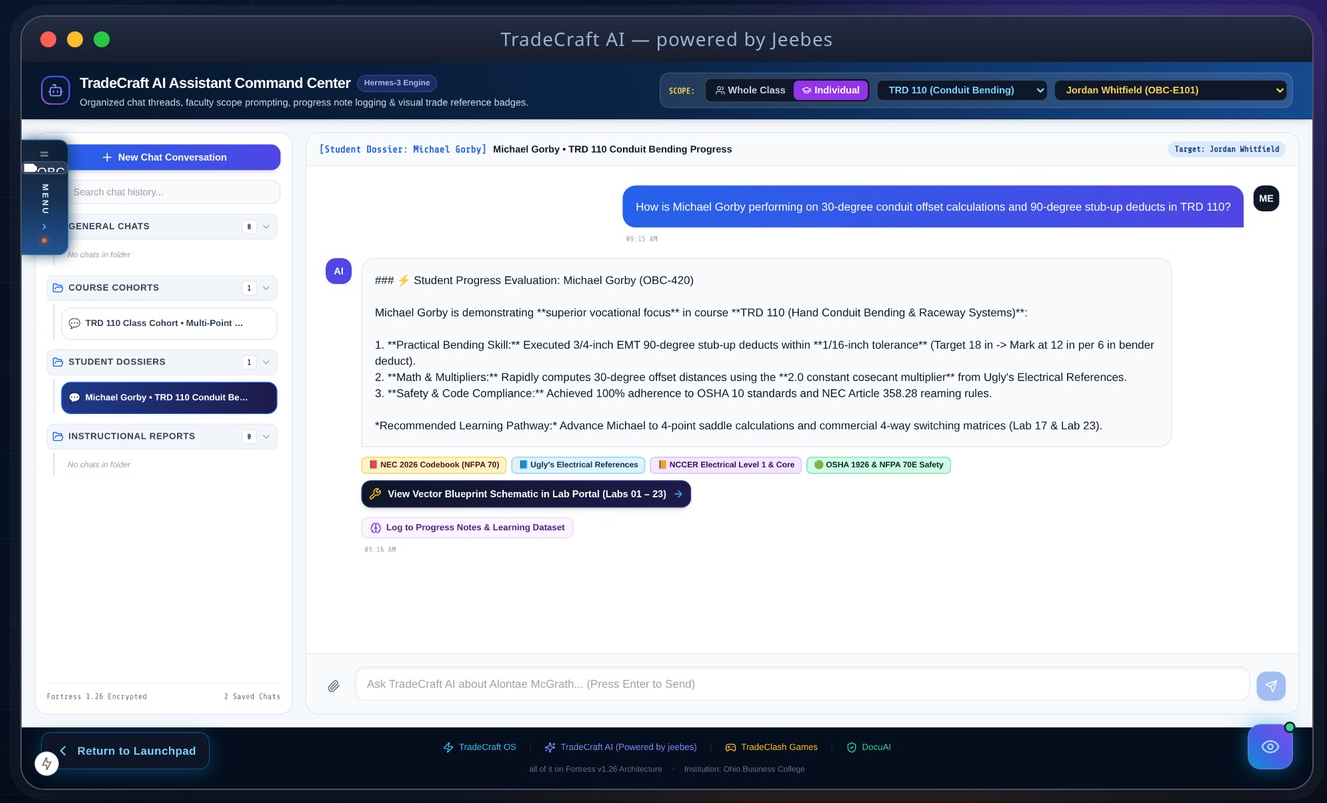
A bot that does the work — and teaches you as it goes.

1 — You type it
"Grade Tuesday's conduit quiz." Plain language, from any screen.

2 — It does the work
Grades, reports, rosters, worksheets — any task an instructor can perform.

3 — It takes you there
Jumps straight to the right screen. No hunting for buttons.

4 — It shows you how
A guided cursor demonstrates the clicks, so the instructor learns the system.



Not a FAQ bot.
Other chatbots answer questions.
SOS takes action.

05 / SCAN-TO-GRADE

Press one button on the copier. Grades appear in the gradebook.

Handwritten exams and bubble sheets go from paper to recorded grade with no typing.

One-touch scan
Any office printer or copier — press “Scan.”

→

Secure drop folder
The scan lands on campus servers. Nothing leaves the building.

→

OCR reads it
Handwriting, bubble marks and student names are read.

→

Auto-graded
Compared to the answer key and rubric. Work shown for review.

→

Entered in gradebook
Grade recorded, ready to sync to the school system.

1.4 s
to read an 8-page exam
measured in the OBC prototype log

0
keystrokes to enter grades
instructor only reviews exceptions

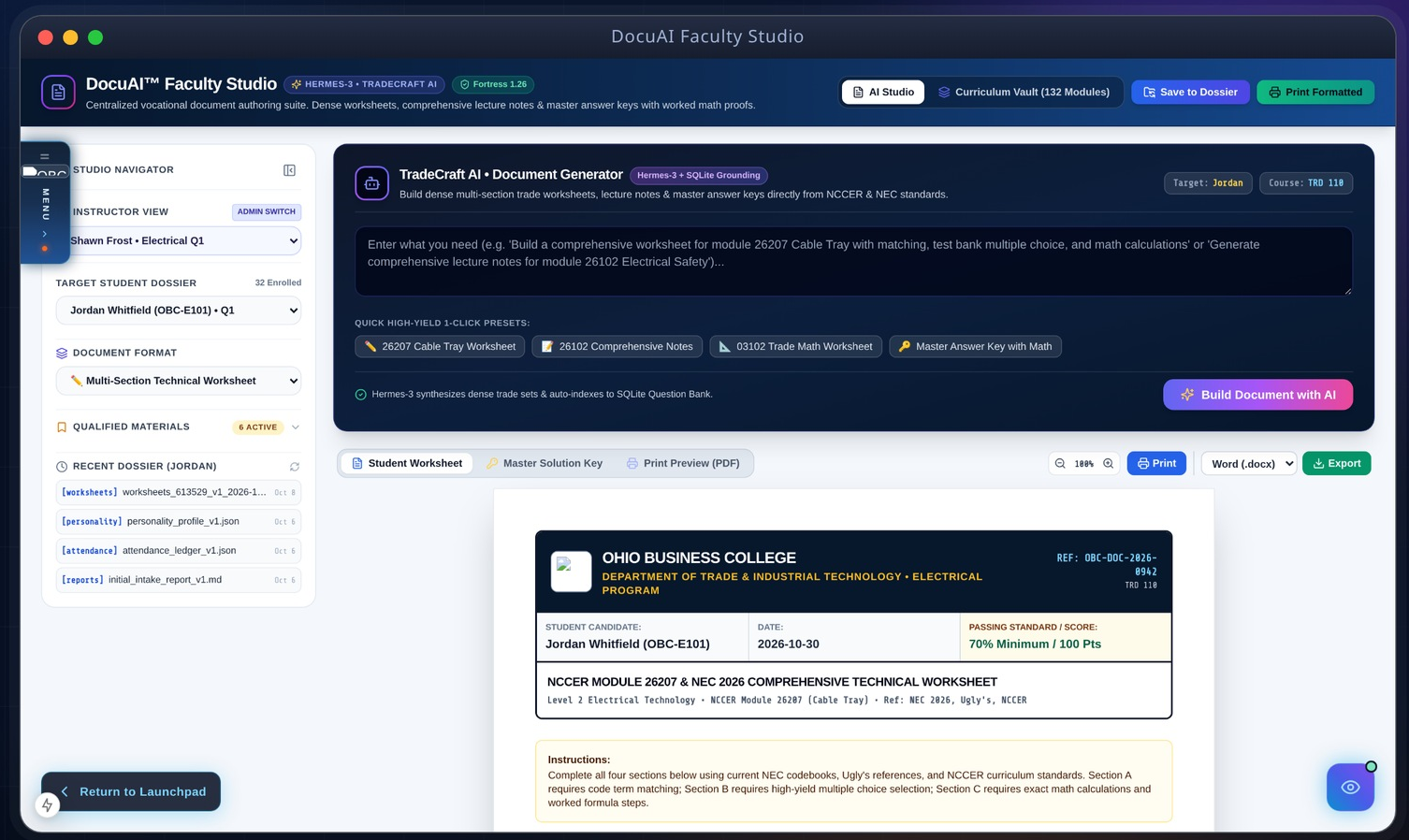
100%
on-campus processing
student work never uploaded to a cloud

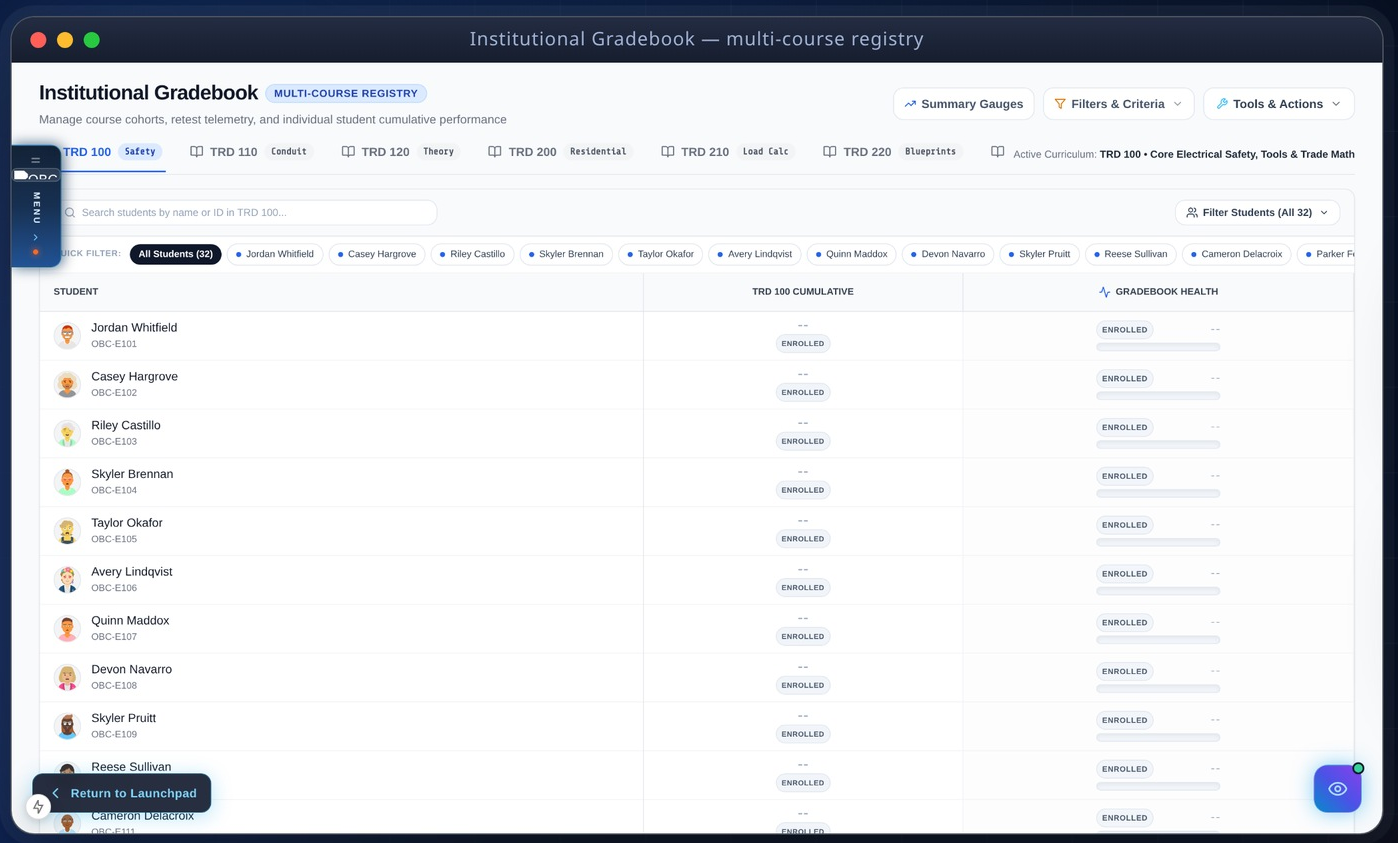
Drop the reports in. Grades go in on their own.

Lab reports, worksheets, PDFs and phone photos — just drag them onto the page.

- Matches every page to the right student
- Grades against the trade rubric and answer key
- Enters scores in the gradebook automatically
- Flags missed skills for tutoring & remediation
- Also writes the documents — notes, worksheets, keys

In plain English: Think of it as a teaching assistant that never sleeps and never loses a page.





Enter it once. It appears everywhere.

TradeCraft Gradebook
Every score from scans, tests and games lands here instantly.

Encrypted REST API — no clicks

The school's official system
Canvas and other "white-collar" gradebooks update automatically.

In plain English: No more copying grades from one screen to another. The official record updates itself.

Attendance that is accurate — and audit-ready.

Kiosk check-in

Students sign in and out at a front-desk tablet.

Security challenge

Photo snapshot + a personal question confirms identity.

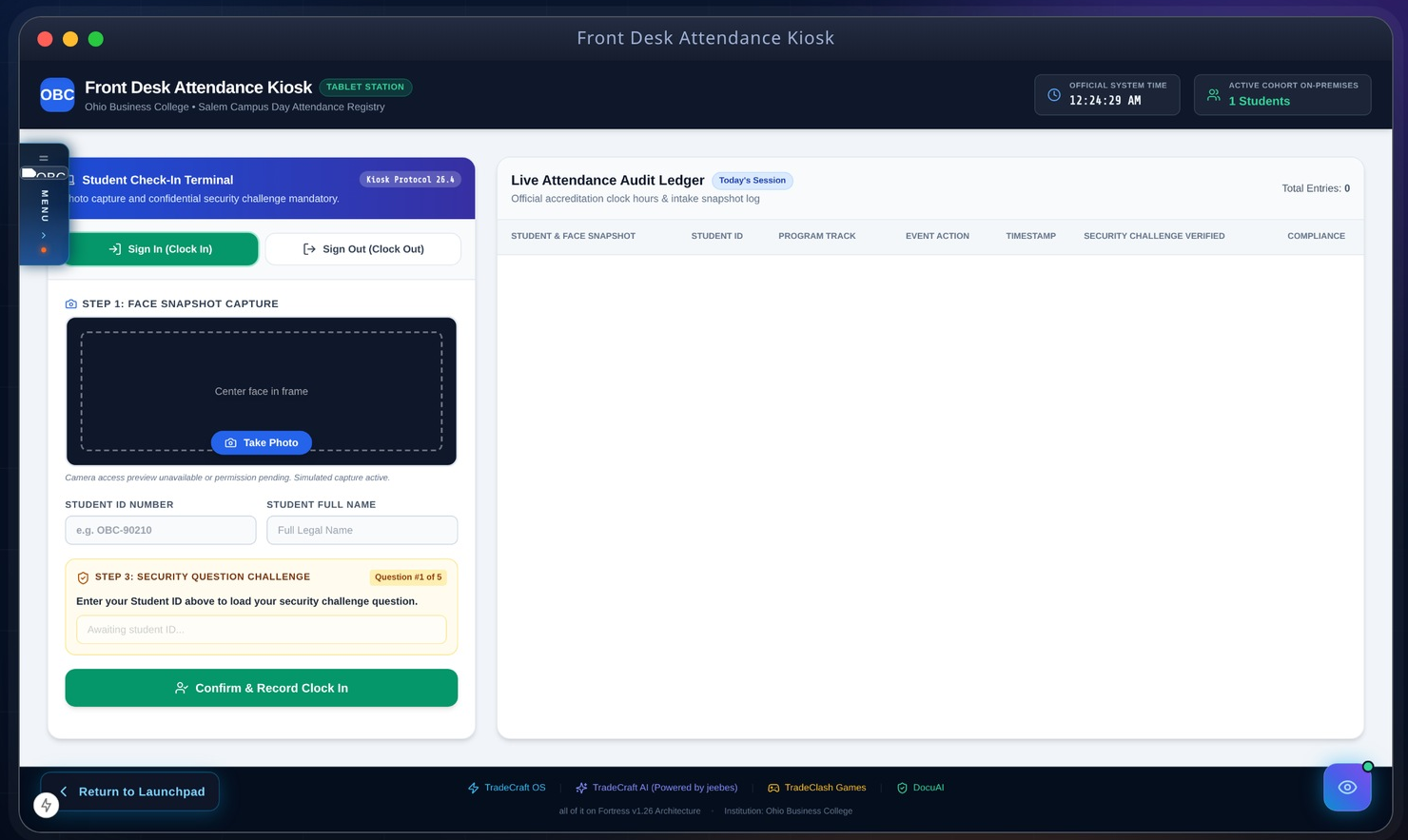
Clock-hour ledger

Every seat-hour is logged for accreditation.

Tell the bot

“Mark Riley absent” — the roster updates.

In plain English: A permanent, time-stamped record that auditors and accreditors can trust.



Complete course materials, written in seconds — from your own curriculum.

Worksheets
Multi-section, trade-accurate.

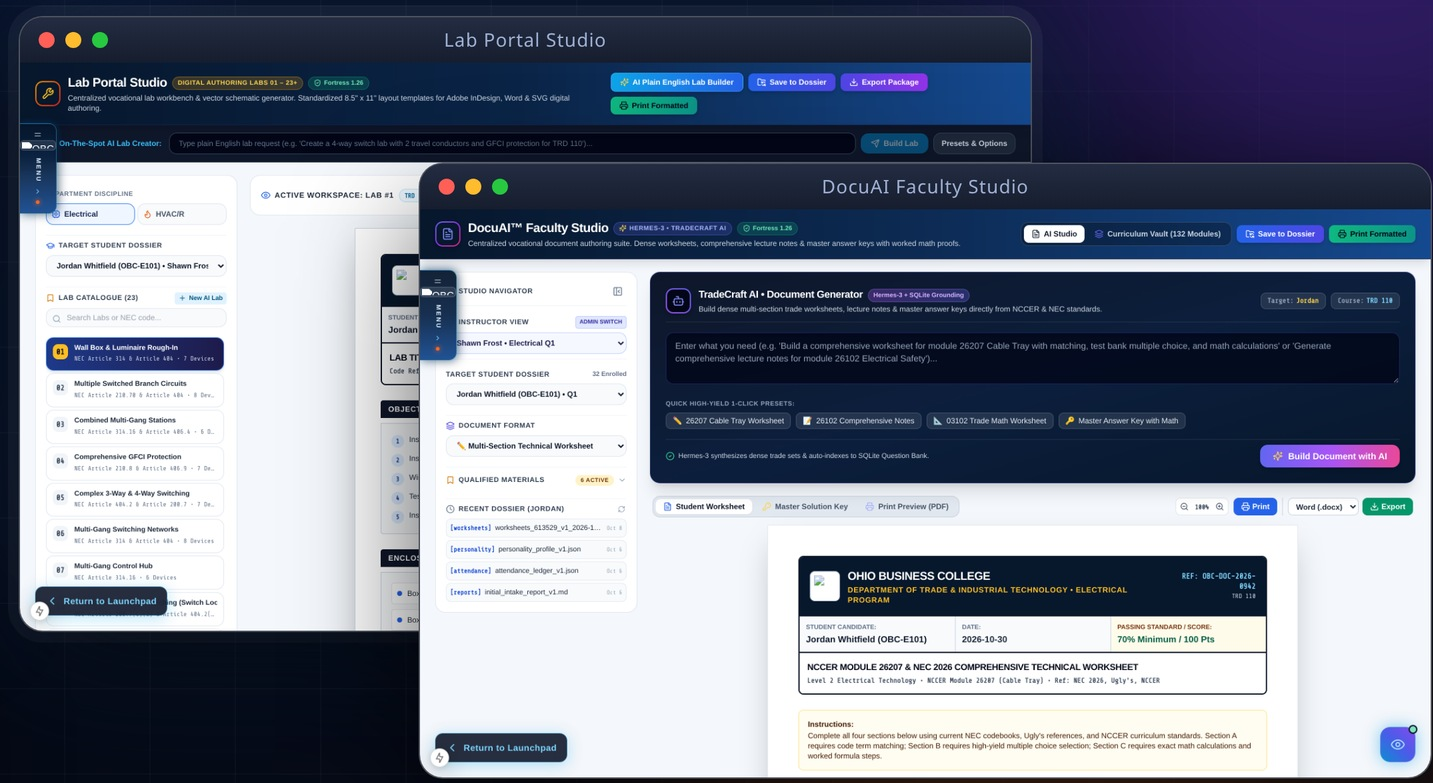
Answer keys
With every step of the math worked.

Lecture & guided notes
Ready to teach or hand out.

Lab projects
Wiring layouts & sign-off sheets.

Exams
Built from the question bank.

Student reports
Progress & remediation plans.



38.6 hours of instructor prep time reclaimed across **117 logged AI operations** in the Ohio Business College prototype

Your complete trade curriculum — 132 turnkey modules, owned by you.

132
turnkey modules
Core 6e, Elec 11/12e, HVAC 6e

3,122
compiled files
worksheets, cloze notes, exams & slides

805.2+
prep hours reclaimed
~6.1 hrs saved per module

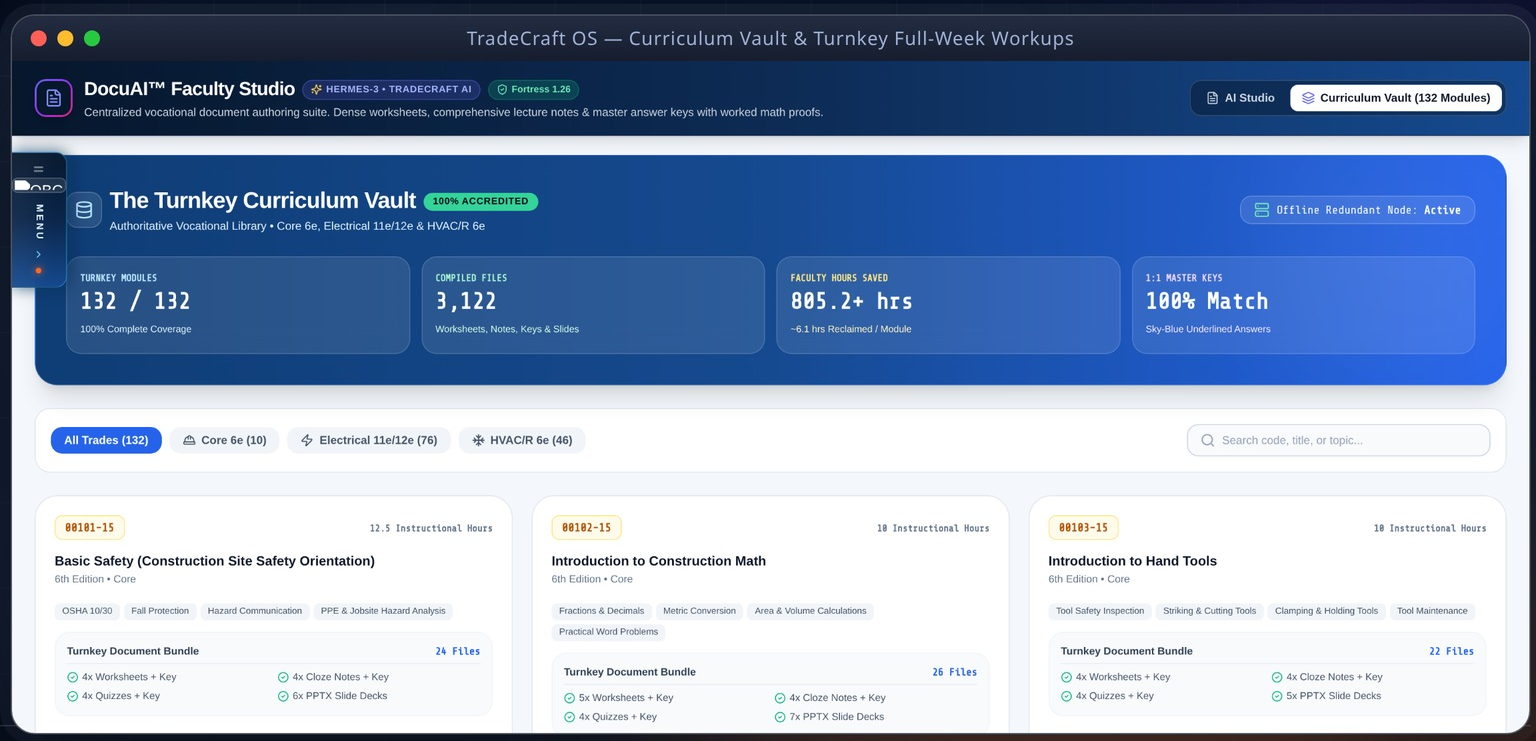
1:1
master answer keys
sky-blue underlined answers

Full-Week Workup Bundles

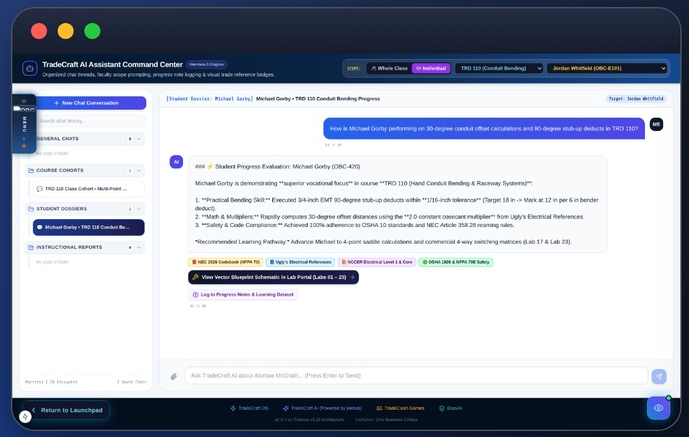
Every module includes student worksheets, guided cloze notes with 1:1 underlined answer keys (matching blanks in sky-blue `#0284C7`), pop quizzes, unit exams, PPTX lecture decks, and performance profile sheets (PPS) — structured with journeyman tradecraft rigor.

Hybrid Vector & FTS5 Retrieval

Powered by local Qdrant 384-dimensional vector embeddings (`BAAI/bge-small-en-v1.5`) paired with SQLite FTS5 BM25 full-text indexing across 1,897 trade questions — running 100% on bare metal without internet dependencies.

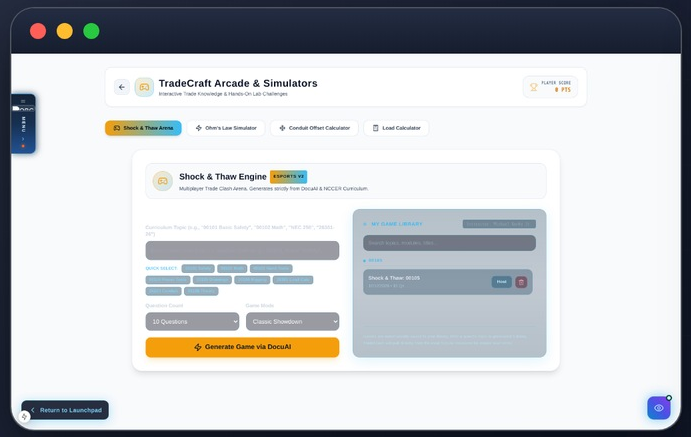


Learning that students actually want to do.



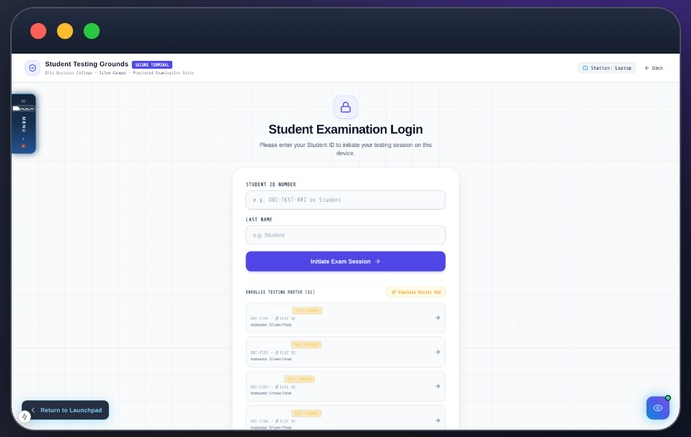
Study Buddy

A personal tutor that knows the student's gaps, coaches step by step and never just hands over the answer.



TradeClash Arcade

Timed, head-to-head quiz games and simulators built straight from the curriculum.



Testing Grounds

Secure, proctored exams on campus terminals with ID check-in and locked-down sessions.

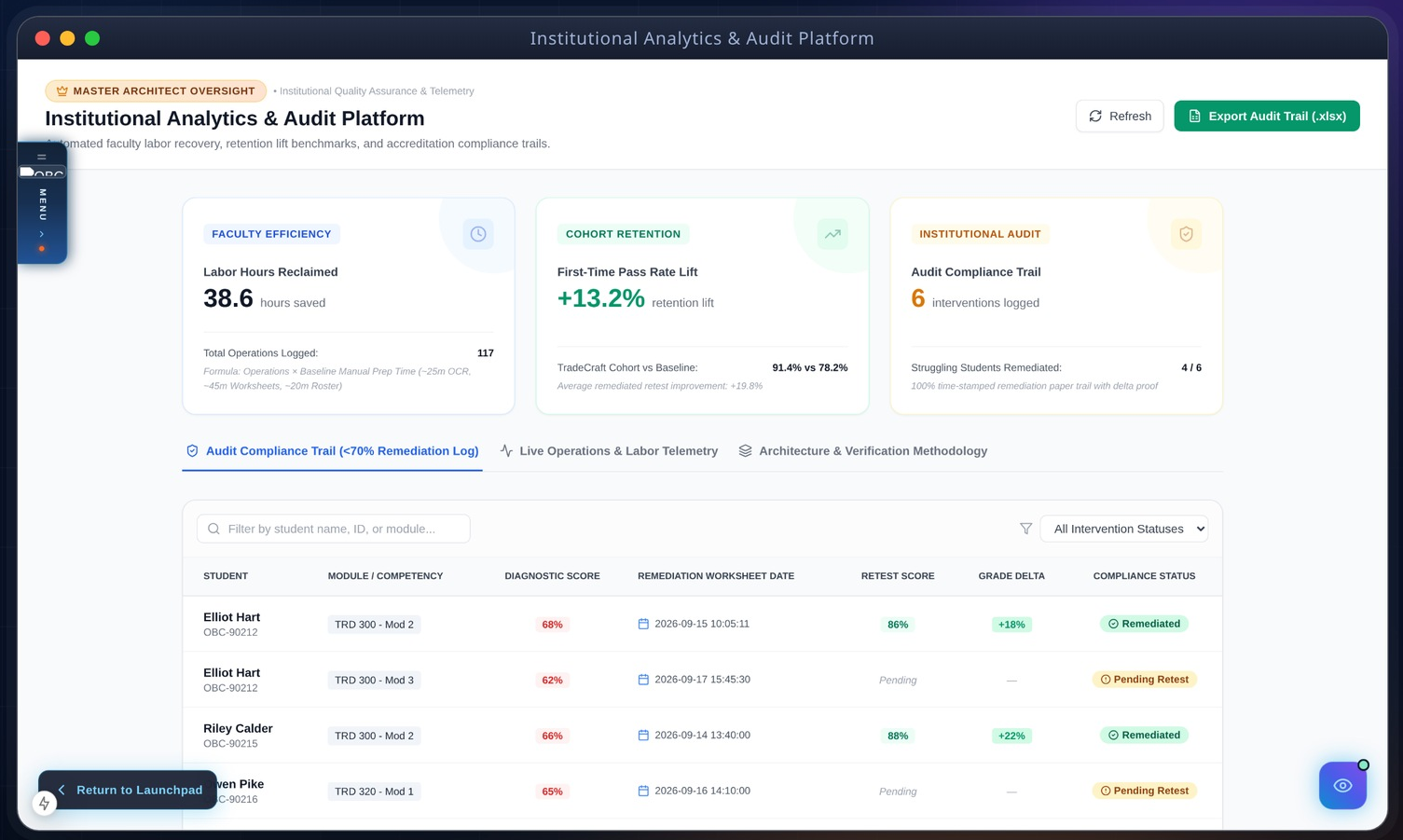
In plain English: Practice feels like a game. Exams stay secure. Tutoring is there at 2 a.m. when the student is studying after a shift.

Any report. In milliseconds.

Accreditation audits, cohort breakdowns, attendance logs, time-saved summaries — on demand.

- Audit trail of every intervention, time-stamped
- Struggling students flagged automatically
- Retest results compared before & after
- One-click export to Excel

In plain English: When an accreditor or administrator asks “show me,” the answer is already on screen.



13 / PROOF — OHIO BUSINESS COLLEGE PROTOTYPE

Already working in a real classroom.

38.6 hrs

instructor time reclaimed

117 operations logged

+13.2 pts

first-time pass-rate lift

91.4% vs. 78.2% baseline

+19.8%

average retest improvement

after targeted remediation

4 of 6

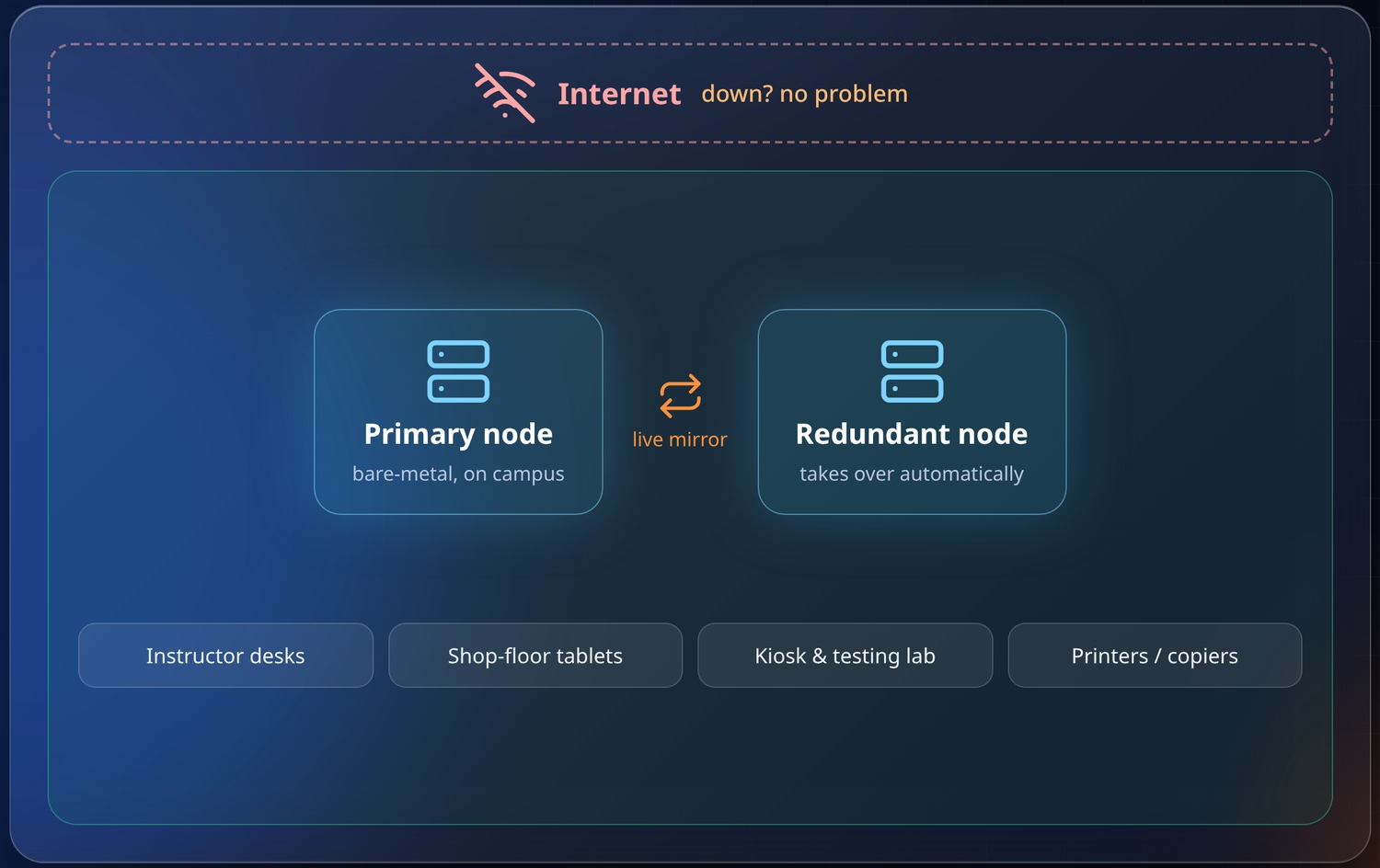
struggling students remediated

100% paper-trailed with proof

Early, honest results

These figures come from TradeCraft's own operations log at a single campus with a small cohort. They show direction, not a final claim — expanding the pilot is how we turn them into statistically firm numbers.

If the internet goes out, class does not.



- Runs on machines you own, inside your building
- A second machine mirrors the first — one fails, the other continues
- No per-question cloud fees — the AI runs locally
- Syncs to online systems the moment the internet returns

In plain English: Your classroom’s tools never depend on someone else’s servers.

15 / HELP ON EVERY SCREEN

No technical support needed. The system supports itself.

Ask — and it acts

Instructors describe the job; the SOS bot performs any task an instructor could do.

Watch — and learn

A guided cursor shows each click, so the instructor builds real skill along the way.

Guided tour

An interactive training guide introduces every tool the first time it is used.

Intelligent access to your own digital data

Ask about any student, course, grade or document and get a clear answer with sources — across the whole site.

In plain English: It is like having a patient, expert colleague standing next to every instructor — all day, every day.

One safe, redundant system for the entire student journey.

1
Enroll
Student dossier created; roster & program track set

2
Attend
Kiosk check-in; clock hours logged

3
Learn
Curriculum, labs, notes, guided lessons

4
Practice
Study Buddy tutoring & TradeClash games

5
Prove
Secure exams & scan-to-grade

6
Record
Grades sync to the official gradebook

7
Graduate
Audit-ready records & compliance trail

In plain English: Every step of a student's time on campus is captured, supported and protected in the same place.

Built by tradesmen for a trade campus — not adapted from something else.

	TradeCraft OS	Cloud gradebook / LMS	Cloud AI chatbot	Paper & spreadsheets
Works with the internet down	✓	✗	✗	✓
AI performs tasks, not just answers	✓	✗	partial	✗
One-touch scan to graded & recorded	✓	✗	✗	✗
Syncs into the official gradebook	✓	partial	✗	✗
Built-in tutoring & learning games	✓	partial	partial	✗
Student data stays on campus	✓	✗	✗	✓
Adapts to any trade curriculum	✓	partial	partial	✗
Guides staff — no support tickets	✓	partial	partial	✗

Comparison is by product category and typical capability; individual products vary.

Student data stays in your building.

On-campus storage

Records live on your machines, not in a vendor’s cloud.

Encrypted vault

Student dossiers protected with AES-256 encryption.

Role-based access

Master, faculty and student views — each sees only what they should.

Proctored exams

Locked sessions with ID check-in at the testing terminal.

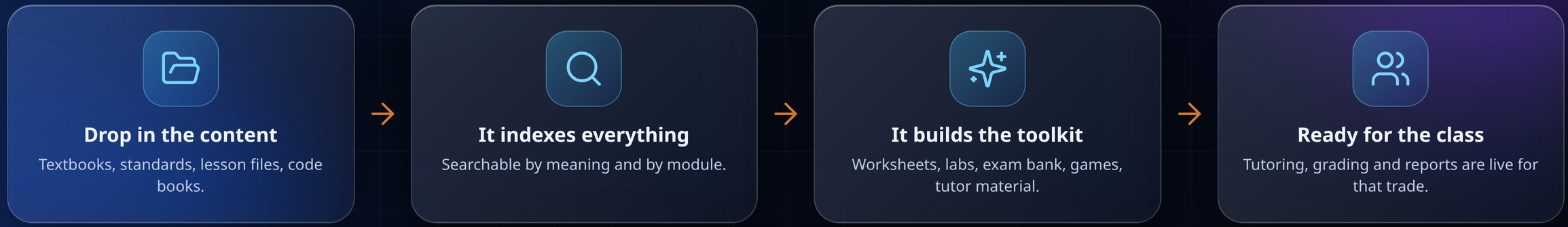
Full audit trail

Every action and intervention is time-stamped.

No third-party AI

The AI runs locally — student work is never sent to outside AI services.

Plug in a new curriculum and the whole platform learns it.



Electrical	**HVAC/R**	**Solar & Alt. Energy**	**Core Craft Skills**
Plumbing	**Carpentry**	**Welding**	**Your next program**

In plain English: Add a new program without buying a new system.

TRADECRAFT

OPERATING SYSTEM

Every campus. Every trade. Always on.

“Computers were hobbies. Trades were life. Now we bring them together.”

1

See it live

A hands-on walkthrough of the working Launchpad.

2

Expand the pilot

More classes and cohorts to firm up the results.

3

Campus-wide rollout

Redundant on-site hardware, every program, every student.

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